

Vons: Compact Decision Models for Frontier-Agent Workflows Technical Report v1.0.1

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Abstract

Vons is a compact decision model for frontier-agent workflows: a frontier model plans and proposes candidates, Vons scores the current state against those candidates and returns a structured decision with an abstention option, and the host alone validates, authorizes and executes. This Technical Report is the engineering record behind the companion paper. It is bound to one hash-verified evidence packet (`publication-evidence-v2.zip`) and describes the architecture and the KASI host boundary, the implementation that exists, the synthetic-pilot results under `vons.metrics/v2`, the shared ONNX bundle and its CPU parity, a four-cell CPU padding decomposition, historical browser timings, the first external evaluation on the original Mind2Web test splits, the local-model suitability screen, provenance and reproduction commands, a failure taxonomy, and the release gates that remain open. Measured findings, each bound to a field-level ledger row (Appendix A): a shared encoder plus both decision heads and required model assets totals 46,001,618 bytes (43.8706 MiB) against the 67,108,864-byte (64 MiB) design target, whereas the two original separate bundles sum to 91,753,063 bytes and do not jointly meet it; 80 finite CPU graph-partition comparisons pass at $\text{atol} = \text{rtol} = 10^{-5}$ with maximum absolute difference 0.0. On the 200-row synthetic test split the Direct head reaches answerable accuracy 1.0 at coverage 1.0 with selective risk 0.09 (covered unanswerable rows count as errors) and ECE 0.115–0.288; because all 182 answerable rows carry the same label `call_tool`, a constant predictor also reaches 1.0, so this accuracy carries no discrimination evidence. The four-step Diffusion head reaches answerable accuracy 0.011–0.060 at coverage 0.11–0.29; the two heads are reported separately and are not ranked. One calibration fit on a separate 200-row calibration split yields coverage 0.91 and selective risk 0.0 on the test split with the temperature at its search lower bound. On Mind2Web (official revision, three held-out splits, $k = 5/10/20/32$) there were 37,512 selection attempts: 31,769 strict 512-token overflows, 5,743 confidence abstentions and 0 accepted selections; candidate recall rises with k to about 0.29–0.32 task-macro at $k = 32$, conditional selection accuracy is 0.0 and complete-case accuracy is undefined. Historical browser measurements of the Direct full graph (500 timed samples per requested provider over five page/session starts) give total-request p50 of 2207 ms (requested WASM) and 91 ms (requested WebGPU); provider execution is unverified, page starts are not independent process launches, and process/GPU memory is null. No completed browser numerical parity matrix exists; the stalled attempt is not a pass. The Jev comparison is not run. Nothing here is a product validation claim.

Licence and status. The report text and its original figures are licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>). The licence does not cover the Vons software, quoted source excerpts, third-party assets, raw benchmark data or private verification records (Section 12). Report type: pilot / negative-result engineering record accompanying the companion paper. Version 1.0.1 (a patch: corrections to existing text, no new measured categories) denotes editorial readiness of the record, not research completion: every research release gate listed in Section 11 remains open exactly as recorded at the evidence freeze. Every measurement was read from one frozen evidence packet and was not re-run for this version (Section 1).

1 Scope, Evidence Basis and Reading Conventions

Authoritative snapshot. Only the frozen archive `publication-evidence-v2.zip` (SHA-256 `f2fb06c2275a60eec6c5b89938123da599b9979bb10d7e65a5667f6ca69152c4`) is used. Its manifest `packet.json` (SHA-256 `7cc5b19cd0e7691919d632d431ff1101d5a5b26a7c11c104c9fa83c132cd5318`; schema `vons.publication-evidence/v2`; frozen 2026-09-24T08:41:25.776377+00:00) lists 131 files; the archive has 133 members, the two additional members being `packet.json` itself and `input-spec.json`, which is covered by the manifest’s `spec_sha256` field. All 131 listed files and all 258 supplied claim rows were verified in the drafting session (131/131 files matched by SHA-256 and byte count; 258/258 claims matched their source field). The verification record accompanies this report as a private file (Section 8). Earlier drafts and registers are preserved and are superseded where they conflict (Section 13).

Code revision (frozen capture state). `packet.json` records `git_revision = null` with `git_revision_reason` “workspace has no Git repository” (`DOCUMENTED STATUS`). This describes the **frozen capture state at the 2026-09-24 freeze**: the workspace from which the packet was frozen was not a Git checkout, so provenance rests on the explicit per-file SHA-256 digests in the manifest and on the bundle-manifest digests for model weights. A repository has since been created by the author (Section 12); no commit of that repository can be bound retroactively to any measurement in the packet.

Input verification is not re-execution. Verifying digests establishes that the reports in the packet are the reports the integrator froze; it does not reproduce their measurements. Nothing in this report was retrained, re-inferred, re-benchmarked or re-run in a browser for any version of the report. The Mind2Web raw archive, normalized rows, per-row predictions and all model weights are deliberately absent from the author packet; their digests and aggregate results are present (Section 6, Section 8).

Labels. Every quantity carries one of the following labels. `MEASURED`: a value transcribed from a named field of a hash-verified file (ledger operation `identity`). `MEASURED (HISTORICAL)`: measured on an earlier configuration that is not the current software. `DERIVED`: an arithmetic combination of measured fields, with the formula recorded in the ledger. `DESIGN TARGET`: an intended property, never an outcome. `DOCUMENTED STATUS`: an integrator-recorded implementation state, not independently executed here. `PLANNED`: intended work, not implemented or not run. `UNAVAILABLE`: not measured, or `null` in the source. `AUTHOR DECISION`: a licensing or status choice made after the evidence was frozen. Values in tables are rounded to the precision shown (at most four decimals; milliseconds to three); the full-precision values are in the named packet fields. Every quantitative statement is bound to a row of the claim ledger that accompanies this report; the body text no longer carries ledger identifiers, which are collected per section and table in Appendix A.

What this report does not claim. Product readiness; external-task quality (Mind2Web selection is a negative result); browser-deployment performance of the current SDK or shared bundle; peak memory; general Diffusion inferiority or superiority; general calibration; verified KASI boundary behaviour on audit logs; a Jev comparison; an adopted open-source licence.

2 Architecture and the Host/KASI Boundary

Decision contract. A Vons request carries a structured state, a bounded question and a candidate list; the model returns a choice with a probability distribution and an answerability/abstention signal. The documented SDK limits are 512 input tokens, 8 questions per request and 32 candidates (`DOCUMENTED STATUS`). Three things about Score questions must be kept apart. (a) The generic contract *schema* admits Score questions with a 2–10 item rubric (`vons/contract.py` validator; `tools/verify_browser_parity.py` accepts the type for request validation only). (b) *Runtime support*: the candidate-selection ONNX head in the browser SDK **rejects** Score questions (`sdk/typescript/src/onnx.ts` throws “score questions are not supported by this candidate-selection ONNX head”), and no Python inference backend in the

frozen snapshot implements Score inference; no Score inference is implemented or measured anywhere in the packet (DOCUMENTED STATUS). (c) An expected-ordinal score and a boolean P(true) mapping are PLANNED evaluation-adaptor behaviour, not current contract outputs. Model-written confidence is never treated as a calibrated probability; calibration is fitted only on a disjoint calibration split (Section 4.4). The Python validator estimates the token budget from characters whereas the TypeScript SDK uses a tokenizer budget, so the two implementations do not yet form one fully harmonized public contract.

KASI is a compatibility layer, separate from the general contract. KASI keeps the five-action public contract—call, clarify, confirm, refuse, respond—on top of Vons’ general choice and abstention machinery. The model may *propose* ordinary calls, clarification, refusal or a response. It cannot grant consent and never executes. The host wrapper requires a host-owned registry of tool names and risk levels, validates tools and arguments, applies confidence thresholds, and **derives confirm from that policy**. The proposal’s risk field is advisory only: it can never lower the registered host risk, and missing or unknown risk fails closed. Consent is bound to the complete call including its arguments through a `KASICall.fingerprint()` SHA-256 (Python) or the canonical `{name, arguments}` key (TypeScript); a bare tool name in `confirmed_tools` cannot authorize a high-risk call. Unregistered tools are refused. The mapping is one-way: KASI calls can be represented as Vons choices, but arbitrary Vons choices do not become tool calls without the host adapter. Risk metadata and the confirmed-call set are therefore **KASI-side host inputs**, not fields of the general decision request. Table 1 summarizes the division of authority.

Verification status. The Python adapter (`vons/kasi.py`) and the TypeScript policy (`sdk/typescript/src/index.ts`) are in the snapshot and are covered by the integrator’s unit tests (241 Python tests with 50 subtests; 10 SDK tests; DOCUMENTED STATUS). No adapter audit log from a real workflow exists in the packet, so boundary metrics—confirm agreement, under- and over-confirmation, tool validity, invariant violations—remain UNAVAILABLE. The boundary is a design requirement whose unit-level behaviour is tested and whose end-to-end behaviour is unverified.

3 Implementation Status

3.1 Components present in the snapshot

Table 2 lists what exists in the frozen source and where the packet evidences it. Every entry is a DOCUMENTED STATUS; none was re-executed for this report.

3.2 Gate status

Table 3 transcribes the integrator’s gate table from `docs/COMPLETION_PLAN.md`. A status entry is not completion evidence. The plan text refers to an intermediate packet digest; this report binds to the v2 packet named in Section 1.

Table 1. Division of authority in the KASI host adapter (DOCUMENTED STATUS: implemented in the frozen source and covered by unit tests; not measured on live workflows).

Decision	Made by	Rule in the frozen source
Propose a call, clarification, refusal or response	Model	Proposal only; never executed by the model
Whether a tool may be used	Host	Unregistered tools are refused
Risk of a tool	Host registry	Advisory proposal risk cannot lower registered risk; missing or unknown risk fails closed
Whether confirmation is needed	Host	<code>confirm</code> is derived from host policy, never asserted by the model
Consent	User, via host	Bound to the complete call including arguments (<code>KASICall.fingerprint()</code> / canonical <code>{name, arguments}</code> key)
Execution	Host	The host is the only execution boundary

Table 2. Components present in the frozen snapshot (DOCUMENTED STATUS).

Area	What exists	Evidence in packet
Shared frozen encoder / tokenizer	google/bert_uncased_L-4_H-256_A-4 at revision 387825ce42db; 512 tokens...	reports/pilot-direct-test-calibrated-v2.json config
Direct head	One-pass candidate scoring; exported ONNX head and full graph	artifacts/pilot/model-manifest.json; browser-direct bundle manifest
Diffusion head	Conditional score denoising; deterministic DDIM, 4 steps inside one graph; 32 fixed slots; host-supplied initial_noise	artifacts/pilot-diffusion/model-manifest.json; shared-v1 heads
Shared-v1 bundle	One encoder + two heads + tokenizer/config/calibration; transformation recorded as “partition existing graphs; no training or parameter changes”	artifacts/shared-v1/manifest.json
Python package vons/	Contract, data, models, training, evaluation (metrics/v2), calibration (calibration/v2), bundle manifest, KASI adapter, Mind2Web importer/evaluator, Ollama benchmark, research packet	13 modules and 19 Python test files under tests/
TypeScript SDK	Contract validation, KASI host policy, explicit WASM/WebGPU capability selection; optional .onnx runtime that verifies the manifest, loads the tokenizer and .onnx.data, and runs both pilot bundles through pinned ONNX Runtime Web 1.30.0 [1]	sdk/typescript/src/{index,onnx}.ts, demo harnesses, 3 TypeScript test files; 10 SDK tests
Tooling	19 scripts under tools/ (Section 8.2)	Source snapshot, hashed
Integrator checks (2026-09-24)	.venv/bin/pytest -q: 241 passed, 50 subtests; ruff exit 0; SDK build exit 0	reports/orchestration-checkpoint-v2.json; not re-executed here

3.3 Not implemented or not measured

The following are absent from the packet and are UNAVAILABLE OR PLANNED: a shared-v1 browser loader; browser Score requests (explicitly unsupported); browser calibration parity; 2- and 8-step Diffusion browser paths; an additional seed matrix in the browser; lifecycle (cancel/dispose/re-initialize) and offline validation; matched padding cells in the browser; any Diffusion browser timing; browser process-level cold start; process or GPU memory; a complete browser numerical parity matrix; the Jev comparison; and a fully harmonized public character-budget contract between the Python estimator and the TypeScript tokenizer budget.

4 Synthetic Pilot

4.1 Data and metric definitions

The test split data/generated/pilot-test.jsonl has 200 rows (SHA-256 e5fdd950a35593b97d9799435a8739177c1668d2ad7d30687546add0bc4de1bc): 182 answerable and 18 unanswerable rows. The calibration

Table 3. Release-gate status recorded at the evidence freeze (DOCUMENTED STATUS; not re-audited).

Gate	Recorded status
G0 bundle manifest	Implemented and verified
G1 real WASM inference	Implemented; parity harness ready; full execution open
G2 browser matrix	Direct WASM/WebGPU controlled cells measured historically; Diffusion cells, fallback browser and lower-resource device open
G3 padding	CPU padding decomposition measured; dynamic redesign optional/open
G4 Mind2Web	Evaluated under evaluation-only authorization; negative result
G5 attribution	Attribution prepared; teacher inclusion open
G6 Jev comparison	Protocol frozen; comparison open
G7 public release	On hold
G8 licence	Proposal only at freeze (see Section 12 for the author decision made afterwards)
G9 Chrome-extension channel	Proposal only

split is a separate 200-row file, `pilot-calibration.jsonl` (SHA-256 `a986ab870f9d54d3905db1f3c71ea4e004922a5bdf43fe822e4c503d7dd3689`), drawn from the same 2,000-row generated set `pilot.jsonl` (SHA-256 `548090beedd52bc18d6a22de26a57efad81c11ddd125def94b63cfcaca9a8ee0`); the split sizes train 1,400 / development 200 / calibration 200 / test 200 are recorded in `artifacts/pilot/model-manifest.json` (MEASURED).

Label composition. Reading the test split directly: all 182 answerable rows are labelled `call_tool`, the 18 unanswerable rows have a null label, and every row has four options (DERIVED from the file). A constant `call_tool` predictor that never abstains therefore reaches answerable accuracy 1.0 and macro-F1 1.0 (a single class), so **synthetic accuracy carries no discrimination evidence**; only coverage, selective risk and the probability-based metrics distinguish behaviours on this split. The scope string in every quality report is “synthetic test split only; not Mind2Web or browser task success”.

Metric definitions. `vons.metrics/v2` corrects two v1 definitions: **selective risk** is errors over covered predictions, where a covered prediction on an unanswerable row is an error (v1 ignored those rows); **Brier** [2] is the standard multiclass score summed over options (range 0–2), whereas v1 divided each row by its option count. **NLL** and **Brier** are means over the *probability population*: answerable rows whose prediction has no error and carries a valid probability vector (`vons/evaluation.py`, `_probability_rows`, which applies `validate_distribution`); this population **includes abstained predictions** when a valid vector exists and therefore differs from the covered population used by ECE. Per row, $\text{NLL} = -\log(\max(p(\text{label}), 1e-12))$ and $\text{Brier} = \text{the sum over the row's options of } (p - 1[\text{option} = \text{label}])^2$, range 0–2; an empty population yields null. **ECE** [3] uses 10 equal-width confidence bins over covered predictions, with covered unanswerable rows counted incorrect. Because ECE is computed over all covered rows including unanswerable ones while answerable accuracy is not, the direction of miscalibration is **not** inferred from ECE alone in this report. The superseded v1 values are listed in Table 16.

4.2 Direct head: three training seeds

Every seed is correct on all 182 answerable rows and covers all 200 rows—exactly the constant-label baseline of Section 4.1, so the 1.0 values establish only that the pipeline runs end to end. The 18 covered unanswerable rows give the uniform selective risk of 0.09 before calibration: the uncalibrated head never abstains. Cross-seed ECE spans 0.1154–0.2881; these are descriptive across saved seeds, not an independent test population.

4.3 Diffusion head: three training seeds (fixed four DDIM steps, 32 slots)

The fixed four-step head abstains on most rows (coverage 0.11–0.29) and is usually wrong when it answers (selective risk 0.810–0.909). Under the v2 definition (multiclass score summed over options, range 0–2) the Brier score is 0.932–0.937; no comparison with the v1 scale is made. The packet

Table 4. Direct head on the 200-row synthetic test split, `vons.metrics/v2` (MEASURED; rounded, full precision in `reports/quality-summary-v2.json`). Because the split is single-label (Section 4.1), answerable accuracy and macro-F1 equal the constant-predictor baseline and are uninformative about learning.

Seed	Ans. acc.	Macro-F1	Coverage	Sel. risk	ECE	Brier	NLL
7	1.0000	1.0000	1.000	0.0900	0.2163	0.0139	0.1053
17	1.0000	1.0000	1.000	0.0900	0.1154	0.0097	0.0871
27	1.0000	1.0000	1.000	0.0900	0.2881	0.0129	0.1013

Table 5. Diffusion head (four DDIM steps [4], 32 slots) on the same test split, `vons.metrics/v2` (MEASURED; rounded). Reported side by side with Table 4 and not ranked against it; no equal-effort tuning record exists.

Seed	Ans. acc.	Macro-F1	Coverage	Sel. risk	ECE	Brier	NLL
7	0.0275	0.0134	0.155	0.8387	0.4483	0.9318	1.8031
17	0.0604	0.0285	0.290	0.8103	0.4602	0.9370	1.8147
27	0.0110	0.0054	0.110	0.9091	0.5088	0.9368	1.8165

Table 6. Calibration fit and calibrated test result for the Direct head, seed 7 (rounded to four decimals; full precision in `reports/pilot-direct-test-calibrated-v2.json`).

Quantity	Value	Label
Calibration split rows / temperature-fit rows / threshold-fit rows	200 / 182 / 200	MEASURED
Target selective risk	0.05	DESIGN TARGET
Fitted temperature (search bounds 0.05–20.0)	0.05, at lower bound	MEASURED
Abstention threshold; risk on calibration split	0.7558; 0.0	MEASURED
NLL before / after (calibration split)	0.1064 / 0.0	MEASURED
Answerability threshold	0.5	MEASURED
Test coverage / selective risk / answerable accuracy	0.91 / 0.0 / 1.0	MEASURED
Test ECE after calibration	0.2233	MEASURED
Abstained test rows	18 (= the unanswerable rows)	MEASURED

records the pilot’s design limitations—pooled-only conditioning, a weak terminal-noise schedule and decoder limitations—and states that the fixed four-step result **does not establish general Diffusion inferiority**. The two heads are reported side by side and are not ranked.

4.4 Calibration (Direct, seed 7; `vons.calibration/v2`)

Calibration first fits a temperature [5] on the calibration split and then fits an abstention threshold on temperature-scaled confidence for a design-target selective risk of 0.05, both on the separate calibration split (Table 6).

Reading. The pipeline runs in the intended order (temperature first, then threshold on temperature-scaled confidence), uses a disjoint calibration split, and on the test split abstains on exactly the 18 unanswerable rows, driving selective risk to 0 at coverage 0.91. The temperature fit saturated at its lower bound, so the temperature is a boundary value rather than an interior optimum, and ECE remains 0.223. This is pipeline evidence on one seed of one head on synthetic data; it supports no general calibration claim. Checkpoint digests are not carried in report configurations, so model identity is bound only by backend, model id, encoder and seed.

4.5 Harness head timings (not deployment latency)

The per-report `latency_p50_ms` values (0.049 ms Direct seed 7; 0.267 ms Diffusion seed 7) time the decision head only, with encoder embeddings precomputed; device and provider are not recorded. They are retained as historical harness fields (`MEASURED (HISTORICAL)`) and are not deployment or bundle latencies.

5 Deployment Artifacts

5.1 Shared-v1 bundle against the 64 MiB model-asset target

The shared bundle is a **partition of the existing graphs with no training or parameter change**; its manifest carries release status “research_candidate; parity and notices required before release”. The two original separate full bundles measure 45,853,779 bytes (Direct) and 45,899,284 bytes (Diffusion) of model assets (`MEASURED (HISTORICAL)`); each passes 64 MiB alone, but their sum, 91,753,063 bytes (`DERIVED`), does not—only the shared bundle makes both heads available within the target. The target governs model assets, not browser runtime, notices, RAM or first download; no licence or notice payload is yet part of shared-v1.

5.2 CPU graph-partition parity (not browser parity)

`reports/shared-v1/parity.json`: provider `CPUExecutionProvider`, ONNX Runtime 1.30.0, 2 intra-op threads, 20 cases giving 80 comparisons (20 Direct; 60 Diffusion over noise seeds 7/17/27), tolerance `atol = rtol = 10-5`, overall pass true; maximum absolute score and answerability differences 0.0 (`MEASURED`).

Table 7. Shared-v1 model-only bundle (MEASURED; runtime and notices excluded). Source: artifacts/shared-v1/manifest.json.

Asset	Role	Bytes	Label
encoder.onnx	encoder graph	553,978	MEASURED
encoder.onnx.data	encoder weights	44,423,168	MEASURED
direct.onnx	Direct graph	22,843	MEASURED
direct.onnx.data	Direct weights	132,096	MEASURED
diffusion.onnx	Diffusion graph	82,287	MEASURED
diffusion.onnx.data	Diffusion weights	74,752	MEASURED
tokenizer/tokenizer.json	tokenizer	711,396	MEASURED
tokenizer/tokenizer_config.json	tokenizer	379	MEASURED
calibration.json	calibration	233	MEASURED
config.json	configuration	486	MEASURED
Model-asset total		46,001,618 (43.8706 MiB)	MEASURED / DERIVED
Design target (64 MiB)		67,108,864	DESIGN TARGET
Size pass (manifest flag)		true	MEASURED
Runtime bytes		null (runtime not included)	UNAVAILABLE

The file’s own scope is “graph partition only; no quality, browser, calibration or latency claim”. This establishes that partitioning preserved the graphs on CPU; it says nothing about browser execution, quality, calibration or latency.

5.3 CPU padding decomposition

reports/onnx-shape-decomposition.json: ONNX Runtime 1.30.0, platform mac05-26.6.2-arm64-arm-64bit, 3 session starts $\times$ 20 repeats $\times$ 4 rows = 240 samples per cell, 5 warm-up requests excluded per session, identical live texts, masks and noise across paired cells (MEASURED). This decomposition supersedes the earlier 20-row bundle timings of Tech Report v0.1, whose report files are not part of the v2 packet (Section 13). Table 8 and Figure 1 give the four matched cells.

Derived comparisons. The comparisons recorded in the file are differences of **inference + readback p50** (cell values 2.416, 14.705, 219.078 and 220.367 ms), not of the total-request column (DERIVED): candidate-slot padding within Direct ($A \rightarrow B$, $4 \rightarrow 32$ slots at 64 tokens) adds 12.289 ms; sequence padding within Direct ($B \rightarrow C$, $64 \rightarrow 512$ tokens at 32 slots) adds 204.373 ms; at the matched 32×512 shape the four-step Diffusion cell exceeds the Direct cell by 1.289 ms ($D - C$). The matched difference is a system comparison at one shape on one CPU, still subject to architectural differences; it is **not** a method-only or DDIM-only cost and is not extrapolated to a dynamic export or to browsers.

Table 8. Four matched CPU cells (MEASURED; ms rounded to three decimals, full precision in reports/onnx-shape-decomposition.json). “Total” is the total request time per row; the derived comparisons in the text use the inference + readback p50 column.

Cell	Configuration	Samples	Total p50 (ms)	Total p95 (ms)	Infer.+readback p50 (ms)
A	Direct, live 4 slots $\times$ 64 tokens	240	2.586	2.711	2.416
B	Direct, 32 slots $\times$ 64 tokens	240	14.930	15.397	14.705
C	Direct, 32 slots $\times$ 512 tokens	240	219.344	232.942	219.078
D	Diffusion (4 DDIM steps), 32 slots $\times$ 512 tokens	240	220.622	232.749	220.367

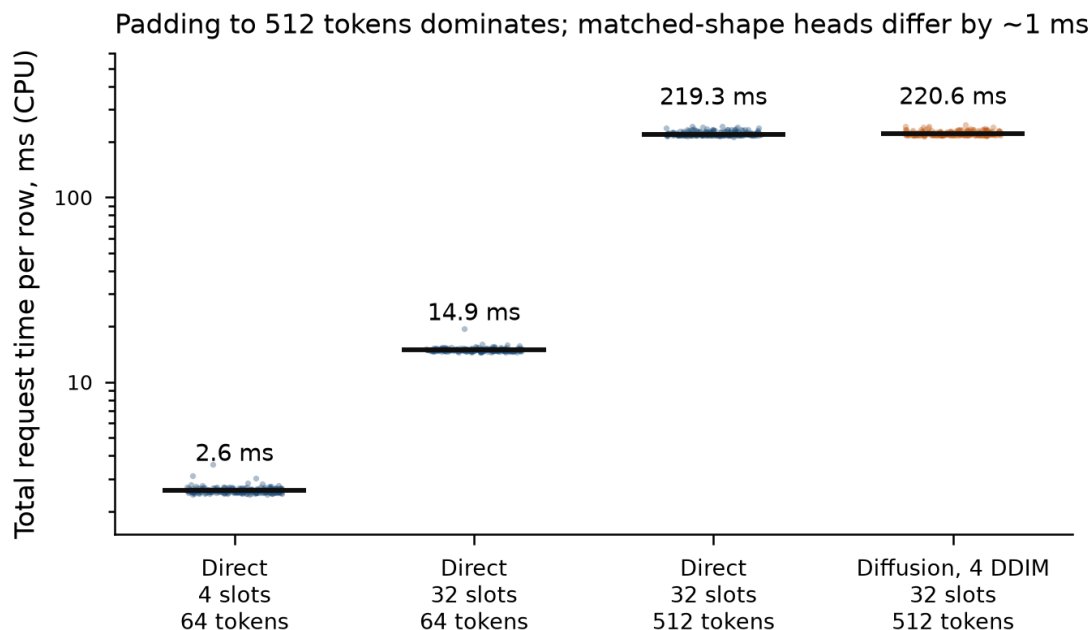


Figure 1. Total request time per row for the four matched CPU cells (log axis; 240 samples each; median marked). Historical full graphs on ONNX Runtime 1.30.0 CPU; not a browser measurement.

5.4 Historical browser measurements (Direct full graph; not the current SDK or shared bundle)

Both runs use the historical Direct full bundle whose manifest digest is 79c7f1e5e7f120e5d9f136fbfb3c6761e1783f73304f10a722b8a71a9966532b (the SHA-256 of artifacts/provenance/browser-direct/bundle-manifest-v1.json), 4 live candidates in 4 slots at 512 tokens, ONNX Runtime Web 1.30.0 [1], Chrome on macOS (user agent recorded), `cross_origin_isolated = false`, WASM threads 1. Each requested provider has 5 page/session starts $\times$ 100 timed requests after 10 excluded warm-ups (Table 9).

Reading rules fixed by the summaries themselves: `requested_is_proven = false`—the harness records only the requested provider and session creation, not graph-partition status, so WebGPU execution is unverified and no speed ratio between the columns is claimed; `process_cold_claimable = false`—“one fresh page/session per source report; browser caches may persist”, so these are not independent browser

Table 9. Historical browser measurements of the Direct full graph (MEASURED (HISTORICAL); ms rounded to three decimals, full precision in the two *-trae-summary.json files). “Requested” means the execution provider was requested by the page, not observed.

Quantity	Requested WASM	Requested WebGPU
Timed samples (successful) / failed	500 / 0	500 / 0
Page/session starts	5	5
Total request p50 (ms)	2207.400	91.400
Total request p95 (ms)	2471.005	132.005
Inference + readback p50 (ms)	2207.000	91.000
Inference + readback p95 (ms)	2470.210	130.460
Tokenize p50 (ms)	0.300	0.200
Requested provider proven	false	false
Process-cold claimable	false	false
JS heap used before / after (bytes; not a peak)	11,757,347 / 11,811,939	57,884,604 / 57,884,604
Process RSS / GPU memory	null	null

process launches; JS heap before/after are two observations, not a peak; process RSS and GPU memory are UNAVAILABLE. Page/session load time ranged 1133–1337 ms across the five WASM page starts. These runs used historical full graphs, not the corrected SDK or the shared-v1 bundle, and they are not evidence of current browser latency.

Browser numerical parity: not established. The internal review record (not public) of the browser-parity attempt records status `incomplete_report_not_recovered`, `correctness_pass` = null and `latency_ms` = null: parity attempts stalled, no completed raw report was recovered, and unit or mocked tests do not substitute. The stalled run is not a pass, and no Diffusion browser timing exists.

6 External Evaluation

6.1 Mind2Web: original test splits, exact-k Direct selection (negative result)

Acquisition. `osunlp/Mind2Web` [6] at dataset revision `17ece8eb89862368edc0cc806acee6fca5163474` (repository revision `33bd95caeee7bba22dd08ecc935845e15c5e5dc7`), licence CC-BY-4.0, usage `evaluation-only`, `no_training` = true, `no_unzipped_redistribution` = true; test archive 567,745,122 bytes, SHA-256 `8f5fbe72afab942fe97cdf7fb397e179885d89b5c16862288e9a14bc6d41ca89`, status `downloaded-and-hash-verified` (MEASURED). Raw and normalized data, retrieval text and per-row predictions stay local; the packet carries their digests and the aggregate reports.

Protocol. Scope per the replay summary: “Replay of saved exact-k action-target predictions over the released candidate pool; no model execution, test-set adaptation, or browser task success.” Prediction scope: “exact-k Direct action-target selection; strict 512-token budget; no calibration; no training”. Retrieval uses the released candidate pool (BM25 [7]) with teacher-forced prior action history—not end-to-end DOM retrieval and not browser task success. Selection uses the uncalibrated shared Direct head with a strict 512-token budget. Bootstrap intervals use 1,000 draws over `task_id_or_example_id`. All 12 expected cells are included (`matrix.complete` = true); Table 10 and Figure 2 show every cell.

Totals. 37,512 attempts; 31,769 strict token-budget overflows recorded as errors; 5,743 confidence abstentions; 0 accepted selections (DERIVED sums over the twelve cells). Because no selection was accepted on any recalled row, conditional selection accuracy and acceptance coverage are 0.0 in every cell and complete-case selection accuracy is null (undefined, not zero). Candidate recall is a property of the released candidate pool at each k, not of the Vons head. The dominant failure is that Mind2Web states exceed the 512-token budget at larger k; the remaining attempts fall below the confidence threshold.

Table 10. All twelve Mind2Web split/k cells (MEASURED; rounded to four decimals, full precision in `reports/mind2web/evaluation-replay-v1/summary.json`). “Overflow” is a strict 512-token budget overflow recorded as an error; “abstain” is a confidence abstention. Conditional selection accuracy (task-macro) is 0.0 in every cell and complete-case selection accuracy is null (undefined) because no selection was accepted.

Split	k	Rows	Positive	Recalled	Recall micro	Recall task-macro [95% CI]	Cond. acc.	Overflow	Abstain	Accepted
Cross-Task	5	2,094	1,975	96	0.0486	0.0713 [0.0549, 0.0894]	0.0	1,246	848	0
	10	2,094	1,975	173	0.0876	0.1199 [0.0983, 0.1429]	0.0	1,867	227	0
	20	2,094	1,975	331	0.1676	0.2100 [0.1833, 0.2397]	0.0	2,081	13	0
	32	2,094	1,975	497	0.2516	0.2916 [0.2623, 0.3223]	0.0	2,092	2	0
Cross-Website	5	1,373	1,314	80	0.0609	0.0806 [0.0615, 0.1025]	0.0	655	718	0
	10	1,373	1,314	128	0.0974	0.1240 [0.0992, 0.1489]	0.0	1,182	191	0
	20	1,373	1,314	234	0.1781	0.2216 [0.1908, 0.2537]	0.0	1,364	9	0
	32	1,373	1,314	361	0.2747	0.3210 [0.2860, 0.3561]	0.0	1,370	3	0
Cross-Domain	5	5,911	5,590	390	0.0698	0.0916 [0.0819, 0.1027]	0.0	3,180	2,731	0
	10	5,911	5,590	648	0.1159	0.1447 [0.1328, 0.1587]	0.0	4,980	931	0
	20	5,911	5,590	1,071	0.1916	0.2248 [0.2099, 0.2405]	0.0	5,852	59	0
	32	5,911	5,590	1,544	0.2762	0.3021 [0.2869, 0.3188]	0.0	5,900	11	0

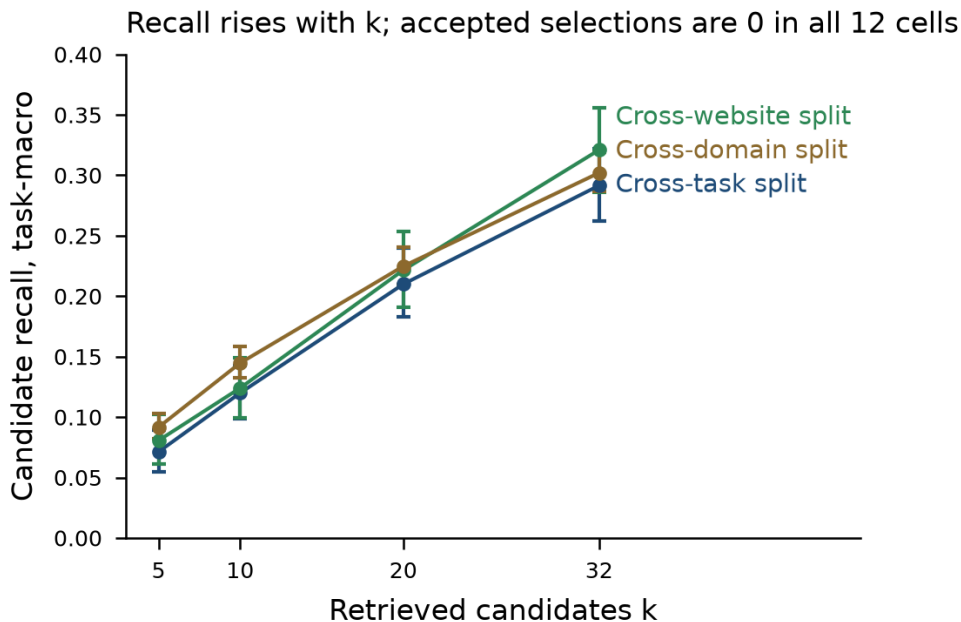


Figure 2. Candidate recall (task-macro, 95% task-bootstrap CI) by k for the three official splits; accepted selections are zero in every cell.

This is a valid bounded negative result for the current contract limits; it is not an end-to-end web-agent evaluation.

6.2 Jev / TypeSafe four-workflow comparison: not run

Status `not_run` with `agreement = null (UNAVAILABLE)`. `docs/JEV_COMPARISON_PROTOCOL.md` freezes the comparison basis (four workflows; Noul/Choice/Score primitives; reference answers averaged from two frontier models; structured wrapper for LLM baselines) and lists the frozen inputs required before any run can be called equivalent. Exact corpus, reference outputs, scorer, provider settings and cost conditions are unavailable; any adaptation to Vons limits must be labelled as adapted. Vendor-published figures [8], [9] are vendor claims and are not reproduced here.

7 Local-Model Suitability Screen and Contributor Attribution

7.1 Screen and alignment audit

All 600 historical responses (200 per model, sequential on one Mac) are in the packet under `artifacts/ollama/`, and `reports/local-model-alignment-v1.json` audits them read-only with `new_model_calls = 0` and `teacher_outputs_used_for_training = false (MEASURED (HISTORICAL))`; Table 11).

The audit’s findings: every answerable pilot label is `call_tool` while all three models selected `respond_directly`; the saved prompts do not identify a required tool or a rule distinguishing those two

Table 11. Historical local-model screen (`MEASURED (HISTORICAL)`); 200 synthetic rows each; no new model calls). Role names are metadata: the three runs share identical prompt sets and one generic system instruction.

Model (local tag)	Role (metadata)	Rows	Known-label acc.	Distinct prompts	Accepted for training
qwen3.8:27b-mlx	writer	200	0.00	10	0
gemma4:31b	judge	200	0.00	10	0
muse-glimmer:30b-mlx	analyst	200	0.09	10	0

actions, so the low accuracies are a **prompt/label-specification limitation, not a measured model error rate**; Qwen and Gemma chose clarify on the 18 underspecified rows where the label requires abstention (a mapping the prompt did not state); Muse’s 0.09 comes from its 18 abstention booleans while keeping a non-null choice, which conflicts with the public abstention contract; role names are metadata, not swapped-role evaluations. No general ranking, role qualification or teacher-data acceptance follows; no local model was admitted as a training teacher.

7.2 Attribution

Models used. Qwen (qwen3.8:27b-mlx), Gemma (gemma4:31b) and Muse Glimmer (muse-glimmer:30b-mlx) were used in writer-, judge- and analyst-role suitability benchmarks; responses are preserved for comparison, not accepted as labels or as calibrated confidence. Muse Glimmer’s official upstream card and LICENSE state Apache-2.0 and the repository also carries a separate usage policy [10]; the local Ollama conversion exposes no licence field and no mapping to an upstream revision, so the **local conversion’s upstream revision remains unverified** and the historical exclusion flag in the raw record is retained as history. Shared encoder/tokenizer: google/bert_uncased_L-4_H-256_A-4 [11], [12] at revision 387825ce42dbb39b87911cdf8e383ee3b25184f8.

Contributors, attributed only to recorded outputs. **Claude Code**—numeric fixes to vons/evaluation.py, vons/calibration.py and their tests (selective-risk denominator, Brier definition, null/finite validation, calibration eligibility), with scoped checks in an internal review record (not public). **TRAE**—tools/summarize_browser_benchmark.py and its tests, and a local licence inventory, accepted with an integrator disposition that grants no redistribution clearance. **Qwen (Ollama)**—one bounded source-only review of the Mind2Web predictor with three findings dispositioned as not reproduced / rejected (existing guard) / accepted diagnostic improvement; the disposition records “One bounded static review, not validated general coding expertise.” **Codex**—integration, shared-bundle export, Mind2Web import/prediction/replay, packet freeze and claim ledger. Writing assistance for this report is disclosed in Section 12.

8 Provenance and Reproduction

8.1 Input verification record

Table 12 records the checks performed on the packet before drafting; Table 13 lists the roles the manifest assigns to the 131 listed files. The full verification record (archive, packet, spec, per-file and per-claim results) is a private file supplied with the report and is not part of the public source.

Table 12. Input verification of the frozen evidence (performed in the drafting session; verification is not re-execution).

Item	Result
Archive publication-evidence-v2.zip	SHA-256 f2fb06c2275a60eec6c5b89938123da599b9979bb10d7e65a5667f6ca69152c4—matches the requested digest
packet.json	SHA-256 7cc5b19cd0e7691919d632d431ff1101d5a5b26a7c11c104c9fa83c132cd5318—matches the requested digest; schema vons.publication-evidence/v2
input-spec.json	SHA-256 matches packet.spec_sha256 (0e464db6cf0c935dbe5818753117c13b01dadbdabcc32de2eeef5841eeb4c31)
Listed files	131 / 131 present with matching SHA-256 and byte count; 0 mismatches. The ZIP has 133 members: the 131 listed files plus packet.json (self; verified by its expected digest) and input-spec.json (covered by spec_sha256)
Claim ledger reports/publication-claims-v2.csv	258 / 258 rows re-resolved (JSON pointer, operation, value, source SHA-256) with 0 failures
Code revision	git_revision = null, git_revision_reason “workspace has no Git repository”—frozen capture state at the 2026-09-24 freeze; no later repository commit is bound to any measurement
Independent re-execution	None; restricted Mind2Web data and model weights are not in the packet

Table 13. Roles assigned by packet.json to the 131 listed files.

Role	Files
source_or_config_snapshot	65
measured_report_or_linked_evidence	14
external_evaluation_aggregate	14
source_or_documentation_snapshot	12
model_or_runtime_provenance_weights_not_included	7
review_or_execution_record_not_independent_truth	7
historical_predictions_metrics_superseded_by_v2	6
historical_local_model_screen_not_accepted_training	3
synthetic_evaluation_input	3

8.2 Reproduction commands (audited against the frozen argparse definitions; not executed)

Every command below was checked against the `add_argument` definitions in the snapshot’s `tools/*.py` and `vons/cli.py`. Angle-bracket tokens are **placeholders** (`<PATH: ...>` a filesystem path, `<NAME>` a string, `<HEX>` a digest); square brackets mark optional arguments; slash-separated lists are the parser’s allowed choices. None of these commands was run for any version of this report.

C1. Recompute `vons.metrics/v2` from saved prediction reports (`--report` is repeatable; source of quality-summary-v2.json).

```
python tools/summarize_quality.py --data <PATH: test jsonl> --report <PATH: prediction report> [--report <PATH> ...]
--output <PATH: summary json> --version v2
```

C2. Temperature-then-threshold calibration on the disjoint calibration split. Recorded paths: `data/generated/pilot-calibration.jsonl`, `reports/pilot-direct-calibration.json`, `data/generated/pilot-test.jsonl`, `reports/pilot-direct-test.json` → `reports/pilot-direct-test-calibrated-v2.json`.

```
python -m vonts.cli calibrate --calibration-input <PATH: calibration jsonl> --calibration-report <PATH: calibration
report> --test-input <PATH: test jsonl> --test-report <PATH: test report> --output <PATH: calibrated report> [--
target-risk 0.05]
```

C3. Partition the two full graphs into one encoder and two heads; writes the shared manifest and, if requested, the CPU parity report.

```
python tools/export_shared_bundle.py --direct <PATH: Direct full graph .onnx> --diffusion <PATH: Diffusion full
graph .onnx> --tokenizer <PATH: tokenizer dir> --output <PATH: new bundle dir> [--direct-calibration <PATH>] [--
parity-report <PATH>]
```

C4. G0 bundle inventory with SHA-256 per asset and the 64 MiB model-asset check.

```
python -m vonts.cli build-bundle-manifest --onnx <PATH> --tokenizer <PATH: dir> --output <PATH: manifest json> [--
checkpoint <PATH>] [--model-manifest <PATH>] [--calibration <PATH> ...] [--runtime-asset <PATH> ...] [--license
<PATH> ...] [--config <PATH> ...] [--source-root <PATH>] [--source-file <PATH> ...]
```

C5. Independent re-enumeration of a bundle manifest.

```
python -m vonts.cli verify-bundle-manifest --manifest <PATH: manifest json>
```

C6. Direct full-bundle versus checkpoint CPU parity.

```
python tools/verify_onnx_bundle.py --checkpoint <PATH: model.pt> --bundle <PATH: full .onnx> --output <PATH: parity
json>
```

C7. Four-cell CPU padding decomposition; the numeric values are those recorded in `onnx-shape-decomposition.json`. The parser defaults differ from the recorded run (session-starts 1, warmup 2, repeats 5).

```
python tools/benchmark_onnx_shapes.py --direct-manifest <PATH> --diffusion-manifest <PATH> --input <PATH: jsonl> --limit 4 --session-starts 3 --warmup 5 --repeats 20 --seed 7 --live-candidates 4 --short-sequence 64 --full-sequence 512 --output <PATH: report json>
```

C8. Evaluation-only import of the pinned official Mind2Web test archive. `--split` takes one or more official split names, or all alone—all cannot be combined with another split (the parser errors otherwise); `--extract-archive` takes a path; restricted data are not in the packet.

```
python tools/import_mind2web.py --input-dir <PATH: acquisition dir> --split <one or more of test_task test_website test_domain, or all alone> --output-dir <PATH: restricted output dir> --source-manifest <PATH: acquisition manifest> [--extract-archive <PATH: test.zip>]
```

C9. Exact-k Direct selections without reading labels. With the default `--backend direct` (the backend of every recorded run) **both `--manifest` and `--manifest-sha256` are required** and the parser errors otherwise (“Direct requires a shared manifest and externally pinned `--manifest-sha256`”); they may be omitted only with `--backend lexical_top1`.

```
python tools/predict_mind2web.py --retriever <PATH: retrieval rows> --output <PATH: predictions> --split <test_task / test_website / test_domain> --k <5 / 10 / 20 / 32> [--backend direct] --manifest <PATH: shared manifest> --manifest-sha256 <HEX>
```

C10. Frozen evaluator for one predictions file.

```
python tools/evaluate_mind2web.py --rows <PATH: gold rows> --predictions <PATH> --output <PATH> [--split <NAME>] [--k 5 10 20 32]
```

C11. Hash-checked replay of the 12 cells (source of `evaluation-replay-v1/`).

```
python tools/replay_mind2web_evaluation.py --rows <PATH: gold rows> --run-index <PATH: run index json> --predictions-dir <PATH> --output-dir <PATH: replay output dir> [--allow-subset]
```

C12. Merge independently saved one-session browser reports.

```
python tools/merge_browser_benchmarks.py --report <PATH: session report> [--report <PATH> ...] --output <PATH: merged json>
```

C13. Validate and recompute a merged browser report.

```
python tools/verify_browser_benchmark.py --report <PATH: merged json> --output <PATH> [--allow-incomplete]
```

C14. Per-session counts and p50/p95 summary (source of the two `*-trae-summary.json` files); exactly one of `--input` or `--report-input` is given.

```
python tools/summarize_browser_benchmark.py (exactly one of --input <PATH: samples jsonl> or --report-input <PATH: merged json>) --output <PATH: summary json>
```

C15. Browser-versus-CPU parity verifier (harness exists; no completed browser matrix report exists).

```
python tools/verify_browser_parity.py --report <PATH: browser trace> --manifest <PATH: bundle manifest> --output <PATH> [--atol 1e-05] [--rtol 1e-05]
```

C16. Read-only alignment audit of the saved local-model responses (`--input` is repeatable, one raw record per model).

```
python tools/audit_local_model_alignment.py --input <PATH: raw ollama record> [--input <PATH> ...] --output <PATH: audit json>
```

C17. Freeze the allowlisted files into a new directory, writing packet.json inside it and a sibling <directory>.zip; refuses to overwrite an existing directory or ZIP.

```
python tools/freeze_publication_packet.py --spec <PATH: input-spec.json> --output <PATH: NEW directory>
```

C18. Derive the field-level claim ledger from the frozen specification.

```
python tools/build_claim_ledger.py --spec <PATH: configs/publication-claims-v2.json> --output <PATH: claims csv>
```

C19. Training, evaluation and export entry points exist with these optional arguments; the exact argument values used for the pilot runs are UNAVAILABLE in the snapshot.

```
python -m vons.cli train / evaluate / export / export-bundle / export-diffusion-bundle [--config <PATH>] [--model <NAME>] [--input <PATH>] [--checkpoint <PATH>] [--output <PATH>]
```

C20. Integrator checks recorded 2026-09-24: 241 tests / 50 subtests, lint exit 0, 10 SDK tests (DOCUMENTED STATUS); not re-executed for this report.

```
.venv/bin/pytest -q; .venv/bin/ruff check .; npm --prefix sdk/typescript test
```

The training/export configurations are configs/pilot.json and configs/pilot-diffusion.json; the argument values used for the recorded pilot runs are UNAVAILABLE in the snapshot. Model weights (model.pt, .onnx.data) are not in the packet; the bundle manifests carry their digests. Historical browser sessions are reproducible only from the merged reports' source-report digests; the raw single-session files are not in the packet.

9 Failure Taxonomy: Current State

The class structure in Table 14 was defined in the earlier private Tech Report sessions and is applied **retrospectively** to the v2 packet; the packet contains no record fixing these classes before its measurements were taken, so the table is a post-hoc classification, not a preregistered one. Boundary and provenance failures are counted with identifiers; behavioural and cost failures may be rates with denominators. A class reports zero only when its instrument ran over a stated denominator.

Table 14. Failure taxonomy applied retrospectively to the v2 packet (post-hoc classification).

Class	Instrument	Current state (v2 packet)
F1 Decision behaviour—contract-invalid output, wrong choice, over/under-abstention, covered unanswerable rows, export parity mismatch	metrics/v2 counts; CPU parity	Direct: 0 wrong answerable choices, 18 covered unanswerable rows per seed (selective risk 0.09) before calibration; 0 after. Diffusion: majority abstention, selective risk 0.81–0.91. Shared-v1 CPU partition parity: 0 mismatches / 80. Browser parity: unavailable (stalled).
F2 Boundary / invariant—model-side consent or execution, model-asserted confirm, unregistered tool executed, fail-open risk, non-deterministic derivation, missing audit record	Adapter unit tests; audit-log replay	Unit-level behaviour tested (241 Python tests, 10 SDK tests, integrator-recorded). Audit-log replay: unavailable —no real-workflow adapter log exists; no class reports zero.
F3 Data and provenance—split leakage, digest mismatch, unflagged synthetic rows, licence/usage-policy unresolved, restricted-data leakage	Packet digests; provenance manifests	131/131 file digests verified; 258/258 claims resolved. Open: Muse local-conversion upstream revision; shared-v1 carries no notice payload; checkpoint digests absent from report configurations. Restricted Mind2Web rows correctly absent from the packet.

Class	Instrument	Current state (v2 packet)
F4 Cost measurement—conflated scopes, unverified provider, non-independent sessions, missing warm-up policy, memory nulls, cross-device carry-over	Timing reports’ own flags	Historical browser: provider unverified (<code>requested_is_proven = false</code>), page starts not process launches, memory null—all disclosed. CPU decomposition: 5 warm-ups excluded, 3 sessions. Earlier 20-row bundle timings superseded.
F5 Harness and analysis—held-out contamination, post-hoc thresholds, best-of-seeds, unavailable rendered as zero	Metric policy; ledger	Calibration split disjoint from test; all three seeds reported; <code>null</code> preserved (complete-case accuracy). No preregistered thresholds exist (open).
F6 Direct/Diffusion asymmetry—unequal tuning, unmatched shapes, different packets	Shape decomposition; training records	Shapes matched at 32×512 on CPU; tuning effort not equalized; Diffusion at one step count. Comparison unadjudicated; no ranking.
F7 External-evaluation validity—budget overflow, retrieval ceiling, teacher forcing, task-success conflation	Mind2Web execution outcomes	31,769 overflows / 37,512 attempts; recall bounded by the released pool; teacher-forced history; no task-success claim.

10 Limitations and Negative Results

1. All quality numbers are synthetic-pilot (200 rows; all 182 answerable rows share the label `call_tool`, so a constant predictor matches Direct’s accuracy) except the Mind2Web negative result; no positive external-task result exists.
2. Mind2Web: 0 accepted selections across 37,512 attempts; the 512-token budget overflows on 31,769; conditional accuracy 0.0; complete-case accuracy undefined. Evaluation-only licence scope; no training or redistribution.
3. Diffusion pilot: pooled-only conditioning, weak terminal-noise schedule and decoder limitations; fixed four steps; no general inferiority or superiority conclusion.
4. Calibration: one seed, one head; temperature at the search lower bound; ECE 0.223 after calibration; no general calibration claim.
5. Browser: historical Direct full graph only; provider unverified; not independent process launches; process/GPU memory null; not the current SDK or shared bundle; no completed parity matrix; Score requests unsupported; Diffusion browser timing absent.
6. The shared bundle’s size result applies to model assets only; notices and runtime are excluded; release status is “research candidate”.
7. No Git revision at the evidence freeze (`git_revision = null`; the freezing workspace was not a Git checkout); a repository created afterwards cannot be bound retroactively to any measurement; checkpoint digests absent from report configurations; historical runtime code identity is bound only where reports record it.
8. Jev comparison not run; the Chrome-extension channel is a proposal.
9. Local-model screen: prompts did not establish the `call_tool` label rule; no ranking, role qualification or teacher acceptance; Muse local revision unverified.
10. Public contract: the Python character-estimate validator and the TypeScript tokenizer budget are not yet a fully harmonized public contract.
11. The integrator’s recorded checks (241 Python tests with 50 subtests; lint exit 0; 10 SDK tests; SDK build exit 0) were not re-run for this report, and unit or mocked tests do not establish browser numerical parity.

11 Release Assessment

Decision at the evidence freeze: **hold** for public release of the software and models as a research/release assessment; the report itself is suitable for publication as a pilot / negative-result engineering

record. The open gates are listed exactly as recorded; version 1.0.1 of this report closes none of them. The author’s separate post-freeze distribution decision is recorded in Section 12; it does not close any gate.

1. Browser numerical parity matrix for Direct and Diffusion on WASM and WebGPU with saved raw reports.
2. Shared-v1 browser loader and calibration parity.
3. Diffusion browser cells, 2/8-step paths, additional seeds, lifecycle/offline and matched padding cells in the browser.
4. Provider-execution evidence and process-level cold start.
5. Memory with explicit scopes or nulls.
6. KASI audit-log replay on real workflow traces.
7. Jev exact inputs or an explicitly adapted benchmark.
8. Preregistered thresholds and analysis plan.
9. Notices inventory for shared-v1 and runtime.
10. Muse local-conversion upstream revision.
11. Licence implementation: at the evidence freeze the free-noncommercial / paid-commercial model was a **proposal only**, the Python package metadata declared Apache-2.0, no new terms were adopted, no open-source claim was made, and upstream conditions were unchanged. The author’s subsequent selection of source-available terms (Section 12) is an `AUTHOR DECISION` whose licence text and notices are not part of the frozen evidence.
12. Cross-review reconciliation of this report against the companion paper (v1.0 consistency check; performed as a separate record).

12 Availability, Licensing and Disclosure

Report. This report and its original figures are licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>), which includes commercial reuse under its terms. The licence does not cover the Vons software, quoted source excerpts or third-party assets, raw benchmark data, or private verification records (`AUTHOR DECISION`; at the v0.2 stage the article terms were still under owner review).

Software and models at the freeze. At the evidence freeze, the Python package metadata declared Apache-2.0 and a proposed alternative licence had not been adopted; the TypeScript package had no licence field, and the shared-v1 bundle carried no licence or notice payload (`MEASURED: pyproject.toml` and the shared-v1 manifest). The research/release assessment of Section 11 (hold; all scientific gates open) is unchanged.

Distribution decision after the freeze (`AUTHOR DECISION`). After the freeze the author decided on **source-only distribution** of the Vons software under custom source-available terms: noncommercial use is free; enterprise and commercial use require a separate agreement; the terms are not an OSI open-source licence; grants already made under the earlier Apache-2.0 terms are not cancelled. Manuscript links are held back pending review. Source publication does not validate model deployment: none of the open gates in Section 11 is closed by publishing source, and no model weights, browser parity or product readiness is implied. The owner personally performs every commit, push and upload; no assistant performs them. Licence text and notices for these terms are not part of the frozen evidence.

Designated destinations. The author has designated <https://github.com/inlevel9-com/Vons>, <https://huggingface.co/INLEVEL9> and <https://huggingface.co/INLEVEL9/Vons> as destinations for the source and for any future model card. Their public-access and upload status was **not verified** from the drafting environment, and this report makes no claim about files present at any of them (`PLANNED`).

Data and records. Mind2Web data were used evaluation-only, were excluded from training and are not redistributed; only aggregate reports and hashes are held in the evidence packet. The release status of the synthetic pilot data is undecided. The frozen evidence packet, the claim ledger and the input-verification record are held by the author; the ledger is available for review on request.

Writing assistance. This Technical Report was drafted by Claude Fable 5.1 in the Tech Report sessions from the verified v2 evidence packet, with cross-review by Claude Opus 5.5 (Section 13, correction pass 2) and a bounded post-review correction pass (v1.0.1) requested by the integrator (Codex). The assistants verified digests, resolved ledger rows, generated the two figures from packet fields and drafted text; they did not generate or alter any measured value, did not train, run inference, benchmark or execute browser tests, and did not modify the paper or any prior artifact. Local models named in Section 7 were subjects of the suitability screen and one bounded code review; none contributed labels or text to this report. The author is responsible for the content, citations and release decisions.

13 Change Log

Each entry states the exact prior location and state of what changed. No measured value, source, model or publication status changed in any correction pass; superseded values appear only here, never in the live results tables.

v0.1 → v0.2 (evidence rebinding, 2026-09-24). Tech Report v0.1 was bound to the earlier research packet (artifacts/research_packet.json, evidence revisions P-1/P-2) and used `vons.metrics/v1`. Version 0.2 rebinds to `publication-evidence-v2.zip` and adds the categories measured for the first time in that packet (shared-v1 bundle and CPU partition parity, CPU padding decomposition, historical browser summaries, Mind2Web, local-model alignment audit). Table 15 lists the corrections with their prior state.

Table 15. Change log: v0.1 → v0.2, the two v0.2 correction passes, v0.2 → v1.0, and the v1.0.1 correction pass.

Version	Prior location and state	Change
v0.2	v0.1 §4 results tables: selective risk, Brier and ECE under <code>vons.metrics/v1</code> (selective risk ignored covered unanswerable rows; Brier divided by option count)	Recomputed under <code>vons.metrics/v2</code> from the same saved predictions; superseded values moved to Table 16
v0.2	v0.1 CPU bundle latency section: 20-row Direct and Diffusion bundle timings from <code>reports/onnx-direct-latency.json</code> and <code>reports/onnx-diffusion-latency.json</code>	Superseded by the four-cell CPU padding decomposition (Section 5.3); the two latency files are not in the v2 packet and their values are not carried
v0.2	v0.1 calibration: threshold fitted before temperature scaling	Replaced by <code>vons.calibration/v2</code> (temperature first, then threshold; Table 6); the earlier threshold value is superseded and is not carried
v0.2	v0.1 §2: risk metadata and confirmed tools described as fields of the decision request	Restated as KASI-side host inputs separate from the general contract (Section 2)
v0.2	v0.1 §2: expected-ordinal Score output described as contract behaviour	Restated as planned evaluation-adaptor behaviour; current SDK returns a discrete choice
v0.2	v0.1 title page: article terms and licence discussed together	Article terms separated from the software licence proposal; both left for owner review at v0.2
v0.2 pass 1	§4.3 sentence “Brier near its maximum”	Removed; measured 0.932–0.937 retained with the v2 range 0–2 stated
v0.2 pass 1	§8.2 reproduction commands as free text	Every command re-audited against frozen <code>add_argument</code> definitions; <code>python</code> prefix on all scripts; explicit placeholders; repeatable flags and choice lists made explicit; <code>freeze --output</code> documented as a new directory
v0.2 pass 1	§9 sentence implying the taxonomy predated the packet	Replaced by an explicit retrospective/post-hoc statement
v0.2 pass 1	Title page and §11: stale “additional authorship” wording	Removed; article terms remained pending
v0.2 pass 2 (R-01)	Abstract, §4.1, §4.2, §10: Direct accuracy 1.0 reported without the constant-label baseline	Stated that all 182 answerable rows are <code>call_tool</code> (read from <code>data/generated/pilot-test.jsonl</code>) so a constant predictor also reaches 1.0; two derived ledger rows added

Version	Prior location and state	Change
v0.2 pass 2 (R-02)	§8.2 <code>predict_mind2web.py</code> command showed <code>--manifest/--manifest-sha256</code> as optional	Both shown as required with the default <code>--backend direct</code> ; parser guard cited
v0.2 pass 2 (R-03)	§5.3 derived comparisons (12.289 / 204.373 / 1.289 ms) presented next to the total-request column	Labelled as inference + readback p50 differences; the four cell p50 values cited; four ledger rows added
v0.2 pass 2 (R-04)	§4.2/§4.3 citation ranges ended before the Brier and ECE rows	Ranges extended to include them
v0.2 pass 2 (R-05)	§3.1 test inventory count	Corrected to 19 Python and 3 TypeScript test files, counted from ZIP members
v0.2 pass 2 (R-06)	§4.1 2,000-row source set stated without a source	Cited to <code>artifacts/pilot/model-manifest.json (/data/rows, /data/splits, /data/sha256)</code>
v0.2 pass 2 (R-07)	§1 and §8.1 archive summary	Corrected to 133 members: 131 listed files plus <code>packet.json</code> and <code>input-spec.json</code>
v0.2 pass 2 (R-08)	§8.2 <code>import_mind2web.py --split</code> shown with a single split	Documented as one or more official splits, or all alone; parser guard cited
v0.2 pass 2 (R-09)	§4 ECE-direction caution	Optional finding not applied; caution retained
v0.2 pass 2 (R-10)	§6.1 recalled column without thousands separators	Separators added; PDF title metadata marked “correction pass 2”
v1.0	v0.2 Markdown manuscript with in-text ledger identifiers and an internal review-record path in §5.4 and §7.2	Rewritten in the house format shared with the companion paper (Typst template <code>vons-template.typ</code>); ledger identifiers moved to Appendix A; the path shown as “internal review record (not public)”
v1.0	v0.2 title page: “Article terms remain for owner review”; §11 item 12	Report text and original figures licensed CC BY 4.0 (AUTHOR DECISION); gate wording updated in Section 11 and Section 12
v1.0	v0.2 §11 item 11: licence “proposal only”	Frozen-evidence state retained verbatim; the author’s post-freeze source-available selection added as an AUTHOR DECISION with the paper’s wording
v1.0	v0.2 title page: “Drafting model: Claude Fable 5.1 ... writing assistance only”; §12 disclosure	Consolidated into the writing-assistance paragraph of Section 12, naming the Opus 5.5 cross-review
v1.0	v0.2 §5.3 and §5.4 tables: milliseconds shown to two decimals (e.g. 2.59, 2207.40)	Shown to three decimals (2.586, 2207.400) with a rounding note; source fields unchanged
v1.0	v0.2 §4.4 and §5.4 tables	Added rows already present in the packet and cited by the companion paper: calibration NLL before/after and answerability threshold; browser tokenize p50
v1.0	v0.2 §10 with ten items	Item 11 (integrator checks not re-run) added from §3.1 for agreement with the paper’s limitations
v1.0	v0.2 had no change-log section	This section and Table 16 added; the v0.1 → v0.2 entries are reconstructed from the v0.2 verification and reconciliation records
v1.0	v0.2 §2 prose only	Table 1 added, mirroring the paper’s authority table; no new property introduced
v1.0.1 (F3)	v1.0 §4.1 “Metric definitions”: NLL and Brier named without their population	Defined as means over answerable rows whose prediction has no error and a valid probability vector (abstained predictions with a valid vector included; differs from ECE’s covered population), with the per-row formulas; bound to <code>vons/evaluation.py</code> lines 39–46, 186–198 and 241–255 (source-code ledger rows added). Values unchanged
v1.0.1 (F4)	v0.2 §2 and v1.0 §2: “Score questions currently return a discrete choice” (the historical v0.1 → v0.2 entry above is kept as history)	Corrected: the contract schema admits Score questions with a 2–10 item rubric (<code>vons/contract.py</code>); the candidate-selection ONNX head rejects them at runtime (<code>sdk/typescript/src/onnx.ts</code> lines 504–506) and no Python inference backend implements Score inference; expected-ordinal / P(true) mappings remain planned
v1.0.1 (F5)	v1.0 §1 “Code revision”, Table 12 row “Code revision”, Limitation 7: “workspace has no Git repository” stated without time	Qualified as the frozen capture state at the 2026-09-24 freeze (<code>packet.json git_revision = null, git_revision_reason</code>); a repository created afterwards cannot be bound retroactively to any measurement
v1.0.1 (F6)	v1.0 §11 “Decision: hold” and §12 “Software and models” paragraph	Hold kept as the research/release assessment at the freeze; a separately labelled post-freeze author decision added: source-only distribution under custom source-available terms, manuscript links held back, source publication does not validate deployment,

Table 16. Superseded v1 and current v2 metric values on the same saved predictions (MEASURED (HISTORICAL) → MEASURED).

Report	Selective risk v1 → v2	Brier v1 → v2	ECE v1 → v2
Direct, seed 7	0.0000 → 0.0900	0.0035 → 0.0139	0.3008 → 0.2163
Direct, seed 17	0.0000 → 0.0900	0.0024 → 0.0097	0.1991 → 0.1154
Direct, seed 27	0.0000 → 0.0900	0.0032 → 0.0129	0.2479 → 0.2881
Diffusion, seed 7	0.8214 → 0.8387	0.2330 → 0.9318	0.4327 → 0.4483
Diffusion, seed 17	0.8000 → 0.8103	0.2343 → 0.9370	0.4537 → 0.4602
Diffusion, seed 27	0.9091 → 0.9091	0.2342 → 0.9368	0.5088 → 0.5088

Version	Prior location and state	Change
v1.0.1	Title, date and running header “v1.0”	owner performs all commits/pushes/uploads; designated destinations named with access status not verified “v1.0.1” (patch: corrections to existing text; no new measured category, value, source, model, licence or publication status change)

Superseded metric values. An earlier metrics version (v1) scored the same saved predictions differently: it ignored covered unanswerable rows in selective risk and divided the Brier score by the option count. Table 16 lists the superseded values next to the v2 values (MEASURED (HISTORICAL)); the v1 values are read from the per-seed report files `reports/pilot-*-test.json`, which the packet lists with role `historical_predictions_metrics_superseded_by_v2`.

A Claim-Ledger Map

The claim ledger `vons_tech_report_v1_0_claims.csv` (492 rows: the 258 supplied v2-* rows, 227 TR2-* rows added in v0.2 for derived values, documented statuses and disclosures, and 7 TR2-* source-code binding rows added in v1.0.1) is a private review file that records, for every row, `source_path`, `source_sha256`, `json_pointer`, `operation`, `value_json`, `scope`, `label` and, new in v1.0, `v1_0_location`. Table 17 groups the rows by the report location that carries them and names the evidence files each group rests on. Paths under the packet’s private review directory are shown as “internal review record (not public)”.

Table 17. Traceability map from report locations to claim-ledger rows and evidence-packet sources.

Location	Ledger rows	Evidence sources
Table 7	V2-001-004	artifacts/shared-v1/manifest.json
Section 5.2	V2-005-009, TR2-028-034	reports/shared-v1/parity.json
Table 4	V2-010-030	reports/quality-summary-v2.json
Table 5	V2-031-051	reports/quality-summary-v2.json
Table 6	V2-052-060, TR2-065-075	reports/pilot-direct-test-calibrated-v2.json
Table 9	V2-061-078	reports/browser-direct-wasm-trae-summary.json; reports/browser-direct-webgpu-trae-summary.json
Table 10	V2-079-258	reports/mind2web/evaluation-replay-v1/summary.json
Section 1; Table 12	TR2-001-007, TR2-224	packet.json; publication-evidence-v2.zip; reports/publication-claims-v2.csv
Table 7; Section 5.1	TR2-008-027	artifacts/shared-v1/manifest.json; artifacts/provenance/browser-direct/bundle-manifest-v1.json; artifacts/provenance/browser-diffusion/bundle-manifest-v1.json
Section 4.1	TR2-035-047, TR2-080-081, TR2-215-219	reports/quality-summary-v2.json; data/generated/pilot-calibration.manifest.json; data/generated/pilot-test.jsonl; artifacts/pilot/model-manifest.json
Section 6.2	TR2-048	reports/quality-summary-v2.json
Section 4.2; Section 4.3	TR2-049-064	reports/quality-summary-v2.json
Table 2; Section 7.2	TR2-076-077	reports/pilot-direct-test-calibrated-v2.json
Section 4.5	TR2-078, TR2-211-212	reports/pilot-direct-test-calibrated-v2.json; reports/quality-summary-v2.json
Section 4.4; Section 10	TR2-079	reports/pilot-direct-test-calibrated-v2.json
Table 8; Section 5.3	TR2-082-104, TR2-220-223	reports/onnx-shape-decomposition.json

Location	Ledger rows	Evidence sources
Table 9; Section 5.4	TR2-105–132	reports/browser-direct-wasm-trae-summary.json; reports/browser-direct-webgpu-trae-summary.json; reports/browser-direct-wasm-merged.json
Section 5.4 (parity not established)	TR2-133–134	internal review record (not public)
Table 10; Section 6.1	TR2-135–153	reports/mind2web/evaluation-replay-v1/ summary.json
Section 6.1 (acquisition)	TR2-154–163	reports/mind2web/data-provenance-v1.json
Section 6.1 (protocol)	TR2-164–165	reports/mind2web/prediction-run-v1.json
Table 11; Section 7.1	TR2-166–192	reports/local-model-alignment-v1.json
Table 2; Section 2; Section 10	TR2-193–198	reports/orchestration-checkpoint-v2.json
Section 5.4; Section 10	TR2-199	reports/orchestration-checkpoint-v2.json
Section 7.2	TR2-200–204, TR2-207	internal review record (not public); docs/ MODEL_ATTRIBUTION.md
Section 11; Section 12	TR2-205	pyproject.toml
Section 2	TR2-206, TR2-209, TR2-232–233	docs/KASI_INTEGRATION.md; docs/ BENCHMARK_PROTOCOL.md; vons/contract.py; tools/ verify_browser_parity.py
Table 3	TR2-208	docs/COMPLETION_PLAN.md
Section 12	TR2-210	drafting-session disclosure
Section 4.3	TR2-213–214	reports/quality-summary-v2.json
Section 8.2	TR2-225–226	tools/predict_mind2web.py; tools/ import_mind2web.py
Table 2	TR2-227	publication-evidence-v2.zip
Section 4.1 (metric definitions)	TR2-228–230	vons/evaluation.py
Section 2; Section 3.3; Section 10	TR2-231	sdk/typescript/src/onnx.ts
Section 1; Table 12; Section 10	TR2-234	packet.json

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